



ARES integration

One contract, then the demonstration. Candidate case study, NEST programme, Open Earth Foundation.

The consortium



multi-year capacity choices



hourly habitat demand



failure and evidence



schema translation

Project Status

Four things have gone wrong at month 8. They are four views of one missing artifact.

| An exchange is two months late, in the wrong format.

Cadence, latency and encoding were never contract terms. An exchange cannot be late against a deadline nobody set.

| Two partners dispute what variables mean and who owns them.

The variables have no declared owner, so the partner that needed names invented them and built against them.

| The risk laboratory says the interface cannot carry its claims.

It has no uncertainty clause. A missing clause, not a complaint about quality.

| Nobody has shown the coupling is necessary.

The operators bridging hourly simulation to multi-year planning are owned by no one.

| Second-year funding depends on a demonstration. This is a crisis to stabilise, not a programme to improve.

Ownership Gap

Read off the partners' own consumes and produces clauses. Nothing inferred.

- GAP-1.

Hourly trace to multi-year bound. Nobody owns the conversion.
- GAP-2.

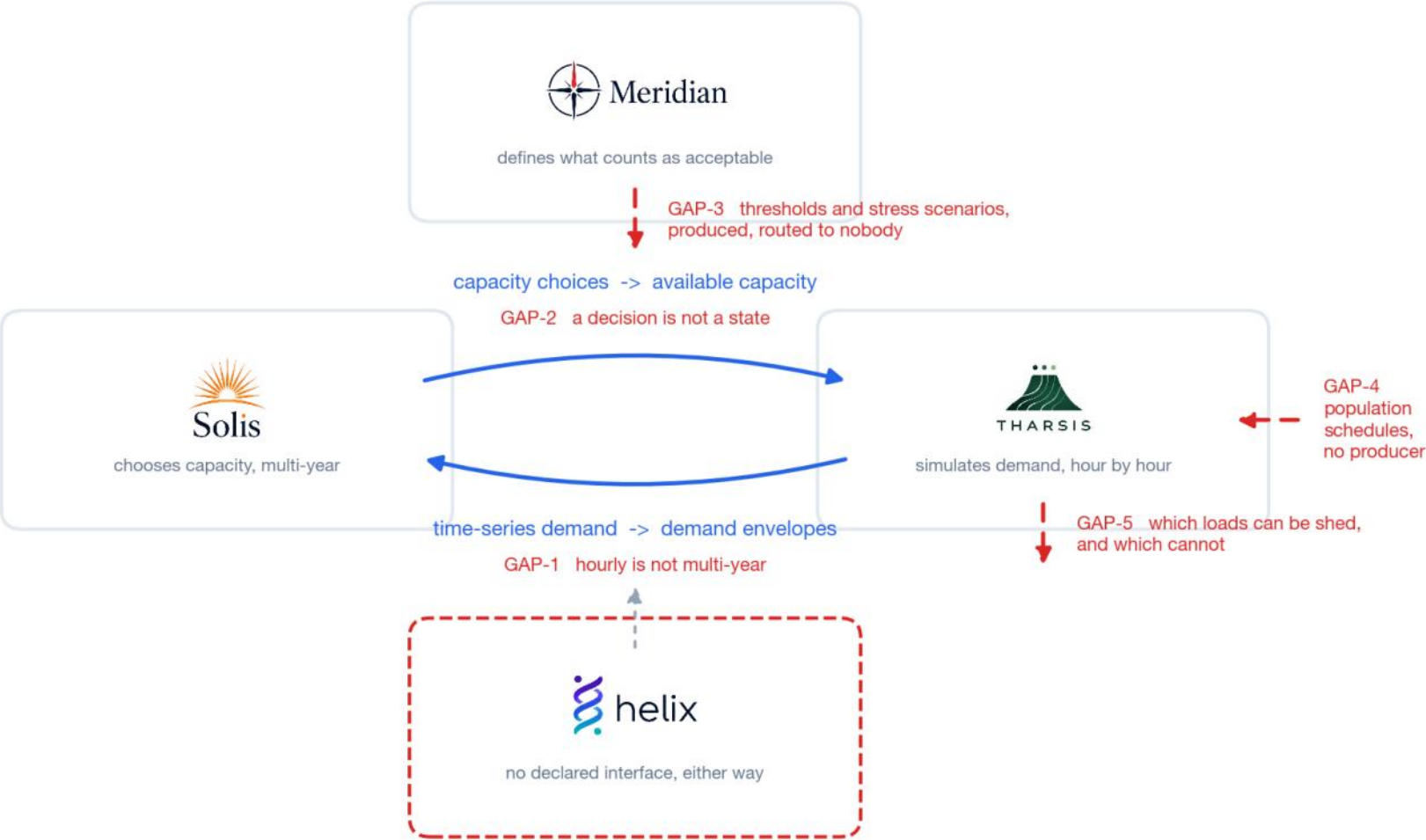
Installed capacity to available capacity. Tharsis derates privately.
- GAP-3.

Thresholds and stress scenarios produced, routed to nobody.
- GAP-4.

The largest demand driver, produced by no partner.
- GAP-5.

Which loads can be shed under stress, and which cannot.

Solid where an exchange has an owner on both sides. Broken where it does not.



Model Fidelity

At what fidelity does this model become admissible evidence for a design decision that kills people if it is wrong? That is the question, not whether coupling is scientifically interesting.

A settlement model exists so somebody can sign a habitat design.

Below a level of resolution and coupling it is not a coarser instrument. It is inadmissible as evidence.

No engineer signs a life-safety design on a tool its own authors would not trust.

That sets a floor, and this consortium is below it.

The physics forces the same answer.

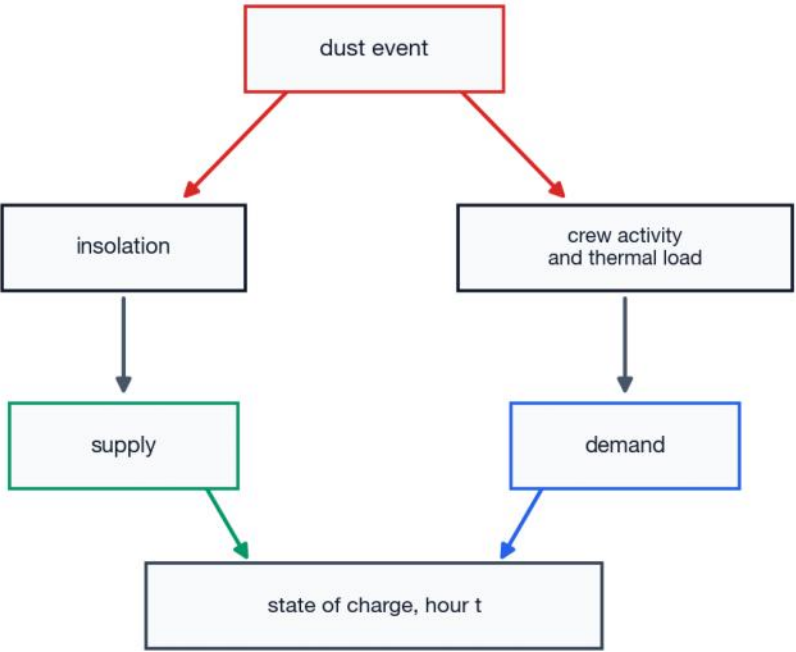
A dust event cuts generation and drives crew indoors at once. Supply and demand answer to one cause.

Exchange bounds once and that dependence is severed by construction.

Meridian's clause 4.2 already forbids it: where two stressors share a physical cause, the model shall not treat them as independent.

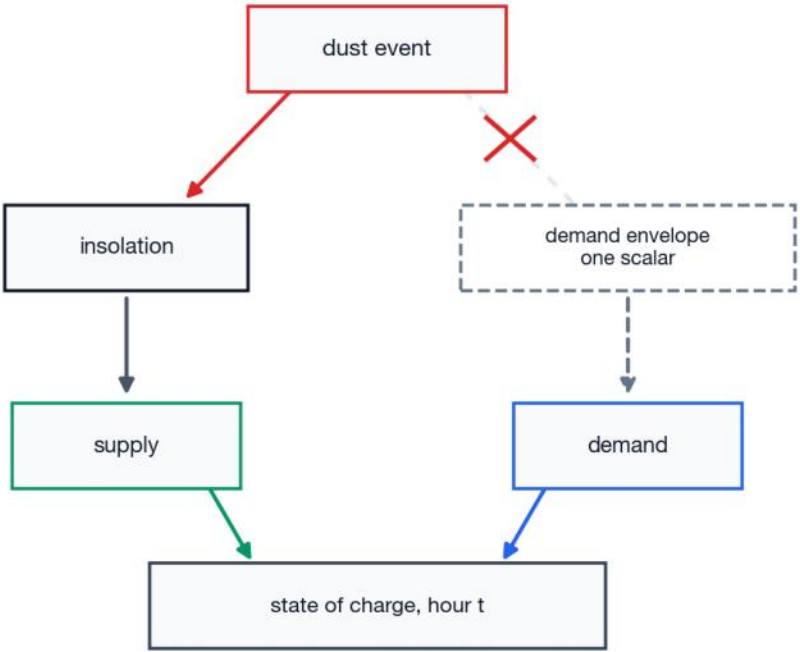
the physics

one cause, two children, dependent by construction



the interface as declared

the arc is severed, so supply and demand are asserted independent



Meridian, clause 4.2: where two stressors share a physical cause, the model shall not treat them as independent.

Delivery Approach

From week 2 the demonstration runs. Every week after improves it rather than building toward a first one.

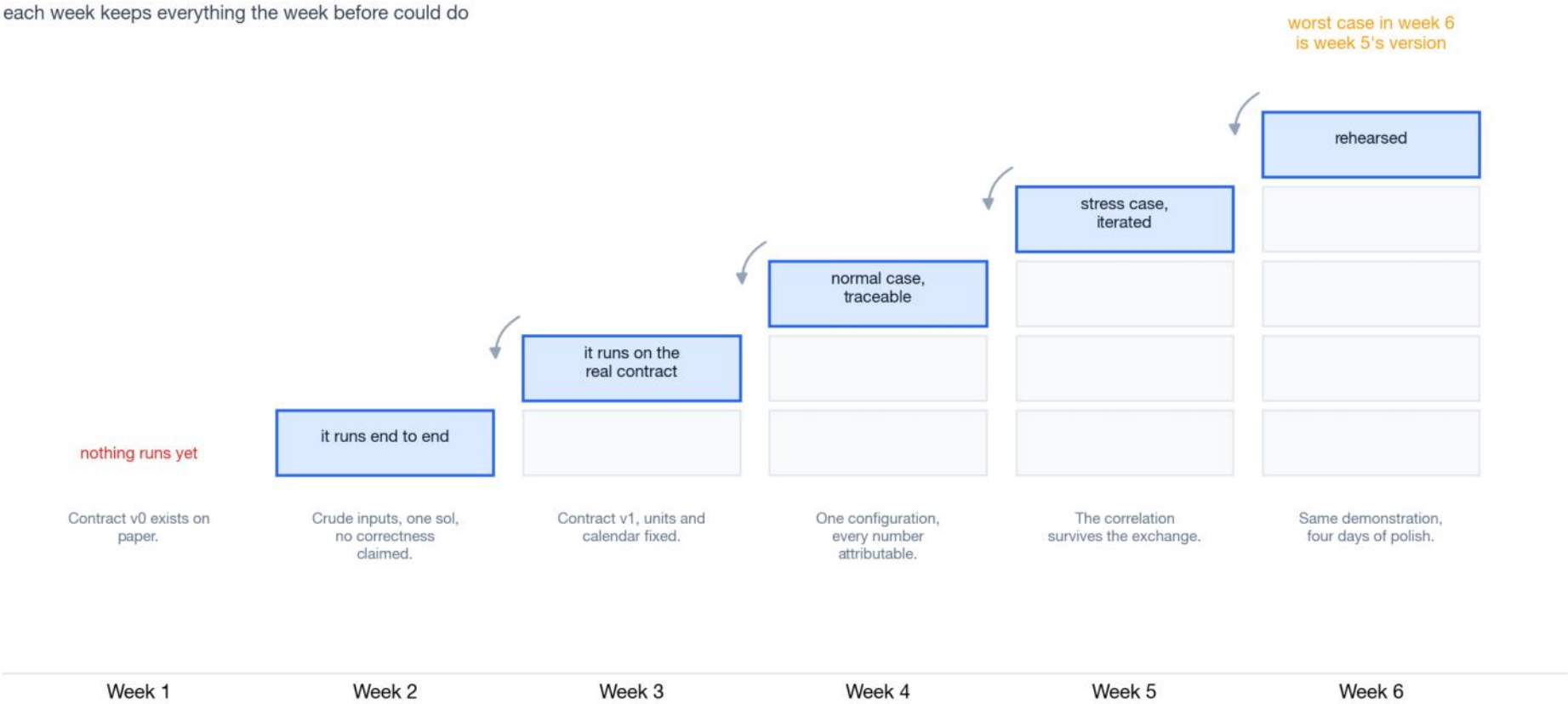
Week 2 runs end to end on crude inputs.
Placeholder numbers, one sol, nothing polished. Its job is to make every format and calendar mismatch fail while there is time.

The consortium is at month 8 having never run end to end. A late exchange in an unexpected format is what that produces.

Each week's output is a superset of the last.
The worst case in week 6 is showing week 5's version.

Week 5 is buffer.
Nothing is scheduled into it that cannot move out.

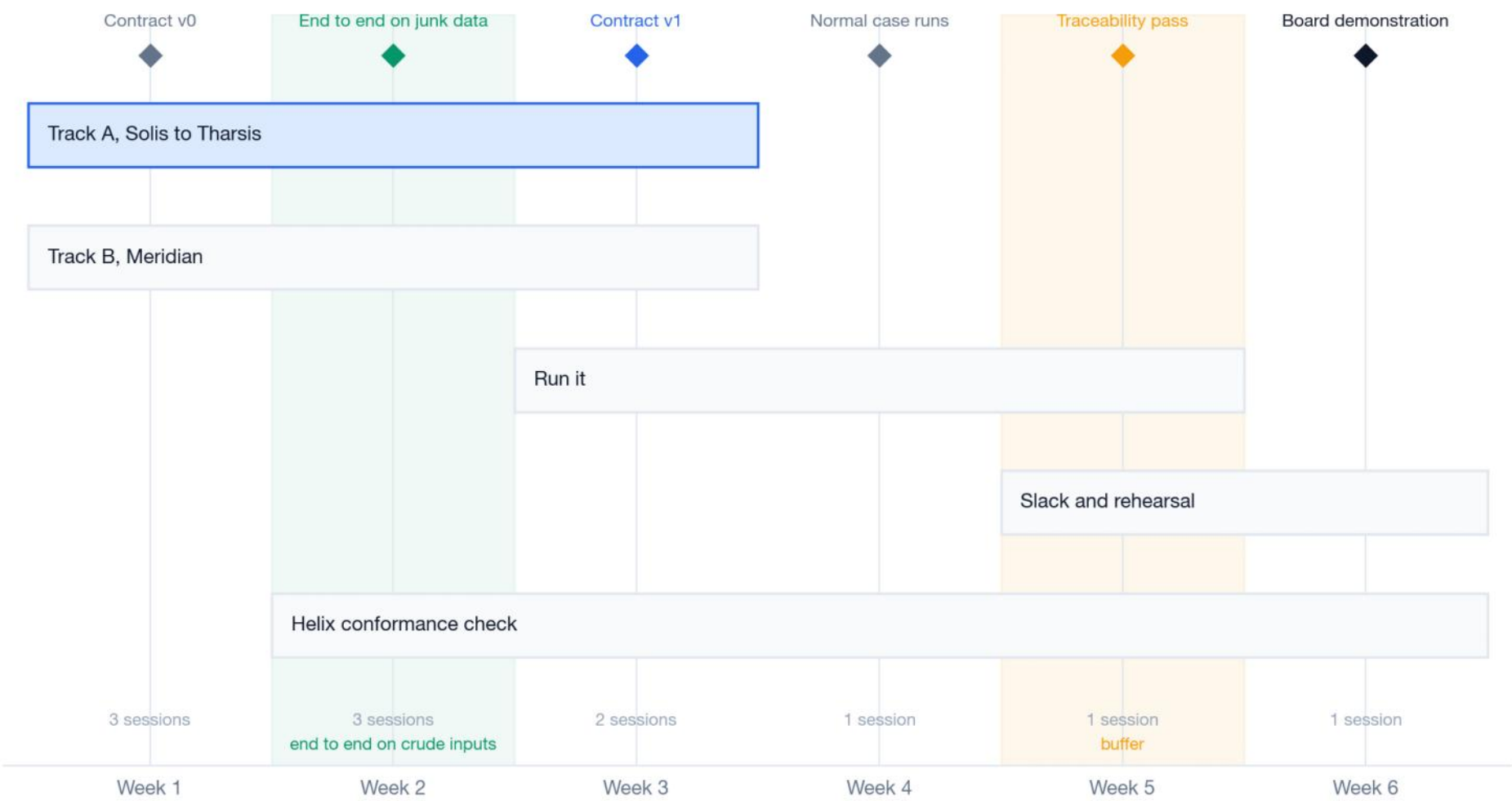
Week 5 also tests the recommendation.
If Solis's revision comes back inside a tolerance set in week 1, the feedback is weak and the coupling argument is wrong.



Six Week Plan

Solis to Tharsis is the critical path. Meridian runs alongside it. Week 5 absorbs a slip and week 6 rehearses.

- Track A is the critical path. The exchange that already failed.
- Track B runs alongside, because what Meridian requires changes what track A must carry.
- Weeks 3 and 4. Normal case, then the stress event run iterated.
- Week 6. The traceability pass. For every number: which model, which contract, which schedule version, which threshold set.
- Every action closes a named gap, so the plan is traceable to the diagnosis.



Governance

Eleven sessions in six weeks. Being present and holding the ruling are different things, and the difference is the governance.

Weeks 1 and 2 carry five of the eleven including the derate session, which needs Solis and Meridian in one room and is the expensive decision in the plan.

Two to three touchpoints a week then a weekly thirty minute standup through week 5.

Helix is in seven of eleven rooms and holds none of the rulings.
It must be present when the contract it built against is settled. It cannot hold the decision.

It also holds the only working transport code for the exchange that failed. We cannot tell from here whether that helps, so the plan gains if it does and loses nothing if it does not.

Filled where the partner is in the room. A dot marks who holds the ruling. Shaded columns settle what a variable means or who owns it.



Partner Alignment

Every partner is worse off under the status quo, and three do not yet know how.

Helix

built against a provisional reading. A declared contract is the only outcome where that work survives. Approach first.

Meridian

has already asked for the uncertainty clause, in the form of an objection. Agree with it in public.

Solis

is judged on a derate function it never saw. Declaring its units is how it takes back control of how its own design is evaluated.

Tharsis

carries blame for thresholds it was never sent and a schedule nobody owns. Declaring the routes moves that liability where it belongs.

In an acute crisis partners accept direction they would decline in business as usual, and that window closes when the crisis is declared over. Week 1 uses it.

Decisions Required

Four decisions and what they buy. Only one of the four costs you anything.

Endorse the diagnosis.

One missing artifact, not four problems. That changes what the next six weeks are for, and it is the decision everything else follows from.

Back the sequence.

Solis to Tharsis first, Meridian alongside it, nothing else started until that exchange works. Saying yes to this means saying no to parallel workstreams that would feel like progress.

Give the coordinator standing to require a declaration.

Units, calendars, owners and what crosses each boundary. Not methods, not code, not anybody's intellectual property. What a partner publishes, not how it computes it.

Accept a narrow demonstration.

One configuration, one stress event, fully traceable. Not a platform, and not four couplings. This is the decision that costs you something, and it is the one that makes week six real.

In return, at week six.

A settlement configuration evaluated under a normal case and a material stress event, with every number traceable to the model, the contract, the schedule version and the threshold set that produced it.